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Part 1: Label and Analyze the Data

Part 2: Train a Model

Part 3: Evaluate Your Model

🕒

Reading: Your Tasks for Part 3

30 min

📋

Graded Assignment: Evaluate Your Model

20 min

Summary and Next Steps

Evaluate Your Model

Assignment details

Due

Mar 10, 11:59 PM IST

Attempts

Unlimited

Submitted

Mar 1, 1:53 AM IST

Your grade

To pass you need at least 100%. We keep your highest score.

100%

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 100%. We keep your highest score.

Next item →

≡ Instructions

1. Calculate each class's average precision (AP) using an IoU threshold of 0.5. Which class has the highest AP?

1 / 1 point

Try again

☐ Accessible

☐ Charger

☒ EV

☐ Correct

We consistently found the EV class to have the highest AP.

2. Calculate each class's average precision (AP) using an IoU threshold of 0.5. Which class has the lowest AP?

1 / 1 point

☒ Accessible

☐ Charger

☐ EV

☐ Correct

The Accessible class consistently had the lowest average precision for our trained models. Did you expect this after analyzing the labeled data?

3. Using an IoU threshold of 0.5, what is the mAP of your detector?

1 / 1 point

0.8765

☒ Correct

Good work. Using the minimum recommended options, we trained models with mAPs above 0.8. Using more optimal settings, our mAP improved to 0.93.

4. Vary the detection threshold and calculate the mAP at each threshold for your model using an IoU of 0.5. Plot the results to estimate an appropriate detection threshold for your detector. Which option below is closest to the value you would choose for your model?

1 / 1 point

Hint: See the EvaluatingObjectDetectionModels.mlx file provided with the course file if you need help doing this.

☐ 0.15

☐ 0.30

☒ 0.55

☐ 0.80

☒ Correct

There is a range of possible values depending on your exact model options. This value was typical for the models we trained.

5. Examine some results by adding the detected bounding boxes to the images. Based on your analysis and examination, what advice would you give for collecting a larger dataset?

1 / 1 point

☒ The model performs fairly well, but you might need more images with the Accessible class to account for the large variance in those signs.

☐ Only EV signs were detected. Consider creating separate models for each class of signs.

☐ Blurry objects are not detected, and you should invest in high-speed cameras to reduce motion blur.

☐ Correct

We agree! This is a good start, but more Accessible signs are needed.

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